

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of BSc Microbiology/ Biotechnology/ Biochemistry Examination April/ May 2019

Course code & Name BS 208 & General Microbiology

Date: 3/5/2019 Day: Friday Time: 10:00 am to 10:30 pm Maximum Marks: 20

## MCQ

### Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

|              |                                                                                                                            |                             |           |
|--------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------|
| <b>Q - I</b> | <b>Choose the correct answer for the following questions.</b>                                                              |                             | <b>20</b> |
| <b>1.</b>    | <b>_____ developed first electron microscope.</b>                                                                          |                             | <b>01</b> |
|              | (i) Jansen                                                                                                                 | (ii) Ruska and Knoll        |           |
|              | (iii) Pasteur                                                                                                              | (iv) None of the above      |           |
| <b>2.</b>    | <b>Streptomycin cannot control growth of _____</b>                                                                         |                             | <b>01</b> |
|              | (i) Gram positive bacteria                                                                                                 | (ii) Gram negative bacteria |           |
|              | (iii) Yeast                                                                                                                | (iv) Both, (i) and (ii)     |           |
| <b>3.</b>    | <b>When you actually observe any object under 10X in a light microscope, the object tends to get _____ fold magnified.</b> |                             | <b>01</b> |
|              | (i) 10                                                                                                                     | (ii) 100                    |           |
|              | (iii) 1000                                                                                                                 | (iv) Can't say              |           |
| <b>4.</b>    | <b>In Gram staining, _____ acts as mordant.</b>                                                                            |                             | <b>01</b> |
|              | (i) Crystal violet                                                                                                         | (ii) Iodine                 |           |

|     |                                                                                                                                          |                                      |    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----|
|     | (iii) Alcohol                                                                                                                            | (iv) Safranin                        |    |
| 5.  | <b>Sand cultures are efficiently used to preserve ____</b>                                                                               |                                      | 01 |
|     | (i) Bacteria                                                                                                                             | (ii) Fungi                           |    |
|     | (iii) Algae                                                                                                                              | (iv) All the above                   |    |
| 6.  | <b>Methylene blue is ____</b>                                                                                                            |                                      | 01 |
|     | (i) Acidic stain                                                                                                                         | (ii) Basic stain                     |    |
|     | (iii) Cationic stain                                                                                                                     | (iv) Both, (ii) and (iii)            |    |
| 7.  | <b>Agar was discovered by ____</b>                                                                                                       |                                      | 01 |
|     | (i) Edward Jenner                                                                                                                        | (ii) Louis Pasteur                   |    |
|     | (iii) Fannie Hesse                                                                                                                       | (iv) Walther Hesse                   |    |
| 8.  | <b>____ uses inorganic compound as a reducing potential/ electron donor.</b>                                                             |                                      | 01 |
|     | (i) Lithotrophs                                                                                                                          | (ii) Heterotrophs                    |    |
|     | (iii) Autotrophs                                                                                                                         | (iv) Phototrophs                     |    |
| 9.  | <b>The limit of resolution of a light microscope is ____</b>                                                                             |                                      | 01 |
|     | (i) 100 nm                                                                                                                               | (ii) 200 nm                          |    |
|     | (iii) 10 nm                                                                                                                              | (iv) 2000 nm                         |    |
| 10. | <b>The time required to reduce the microbial population by 90% at a specified temperature and specified conditions is called as ____</b> |                                      | 01 |
|     | (i) D-value                                                                                                                              | (ii) F-value                         |    |
|     | (iii) Z-value                                                                                                                            | (iv) None of the above               |    |
| 11. | <b>____ is sterilization by moist heat under pressure.</b>                                                                               |                                      | 01 |
|     | (i) Hot Air Oven                                                                                                                         | (ii) Incineration                    |    |
|     | (iii) Filtration                                                                                                                         | (iv) Autoclave                       |    |
| 12. | <b>Bacteria growing at high temperature are called as ____</b>                                                                           |                                      | 01 |
|     | (i) Thermophiles                                                                                                                         | (ii) Barophiles                      |    |
|     | (iii) Halophiles                                                                                                                         | (iv) Neutrophiles                    |    |
| 13. | <b>Bacteria produce antibiotics during ____ phase.</b>                                                                                   |                                      | 01 |
|     | (i) Lag                                                                                                                                  | (ii) Log                             |    |
|     | (iii) Stationary                                                                                                                         | (iv) Death                           |    |
| 14. | <b>____ is not true about Prokaryotes.</b>                                                                                               |                                      | 01 |
|     | (i) Lacks nucleus                                                                                                                        | (ii) Lacks membrane bound organelles |    |
|     | (iii) Possess 70S ribosomes                                                                                                              | (iv) Possess 80S ribosomes           |    |

|     |                                                                                                                     |                          |    |
|-----|---------------------------------------------------------------------------------------------------------------------|--------------------------|----|
| 15. | <b>Penicillin is _____ antibiotics.</b>                                                                             |                          | 01 |
|     | (i) Narrow spectrum                                                                                                 | (ii) Broad spectrum      |    |
|     | (iii) Cell wall inhibiting                                                                                          | (iv) Both, (i) and (iii) |    |
| 16. | <b>_____ technique can always differentiate between live and dead cells during measurement of microbial growth.</b> |                          | 01 |
|     | (i) Taking OD of culture                                                                                            | (ii) Viable count        |    |
|     | (iii) Direct Microscopy                                                                                             | (iv) None of the above   |    |
| 17. | <b>Bleach is chemically _____</b>                                                                                   |                          | 01 |
|     | (i) Phenol                                                                                                          | (ii) Alcohol             |    |
|     | (iii) Halogen                                                                                                       | (iv) Surfactants         |    |
| 18. | <b>CJD is caused by _____</b>                                                                                       |                          | 01 |
|     | (i) Bacteria                                                                                                        | (ii) Viruses             |    |
|     | (iii) Prions                                                                                                        | (iv) Fungi               |    |
| 19. | <b>Ethylene oxide is _____</b>                                                                                      |                          | 01 |
|     | (i) Surfactant                                                                                                      | (ii) Sterilizing gas     |    |
|     | (iii) Oxygen free radical                                                                                           | (iv) Heavy metal         |    |
| 20. | <b>_____ are the beneficial microbes for humans.</b>                                                                |                          | 01 |
|     | (i) Antibiotics                                                                                                     | (ii) Prebiotics          |    |
|     | (iii) Probiotics                                                                                                    | (iv) Synbiotics          |    |

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of BSc Microbiology/ Biotechnology/ Biochemistry Examination April/ May 2019

Course code & Name BS 208 & General Microbiology

Date: 3/5/2019 Day: Friday Time: 10:31 am to 1:00 pm Maximum Marks: 50

## Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

| SECTION – I                                                 |                                                                                                    |    |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----|
| Q - II Answer the following questions as directed. (Any 10) |                                                                                                    | 20 |
| 1                                                           | Define: Microbiology                                                                               | 02 |
| 2                                                           | What is the significance of phenol coefficient test?                                               | 02 |
| 3                                                           | Short note: Lyophilization                                                                         | 02 |
| 4                                                           | Highlight contribution of Louis Pasteur.                                                           | 02 |
| 5                                                           | Define: Sterilization and Disinfection                                                             | 02 |
| 6                                                           | Define: Auxochrome and Chromophore groups in a stain                                               | 02 |
| 7                                                           | Explain: growth rate and generation time                                                           | 02 |
| 8                                                           | Explain significance of chemostat and Turbidostat?                                                 | 02 |
| 9                                                           | Classify: Microorganisms based on oxygen requirements                                              | 02 |
| 10                                                          | What is diauxic growth curve?                                                                      | 02 |
| 11                                                          | Explain radiation to control microbial growth                                                      | 02 |
| SECTION – II                                                |                                                                                                    |    |
| Q-III Answer the following questions as directed. (Any 6)   |                                                                                                    | 30 |
| 1                                                           | Correlate between numerical aperture, resolution and limits of resolution.                         | 05 |
| 2                                                           | Comment on Whittaker's five kingdom classification                                                 | 05 |
| 3                                                           | Draw and signify each phase of the Microbial growth curve                                          | 05 |
| 4                                                           | Elucidate: Heat as a physical agent to control microbial growth.                                   | 05 |
| 5                                                           | Define: Enrichment. Highlight various chemical methods used to achieve the same in the laboratory. | 05 |
| 6                                                           | Compare and Contrast: Cellular versus Acellular life                                               | 05 |
| 7                                                           | Differentiate: Gram positive and Gram negative bacteria                                            | 05 |

